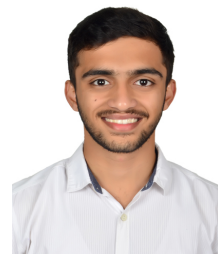


# Shreyas Phansalkar

Vidyadhan Apartment, Erandwane, Pune, Maharashtra

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## Objective

I am an Electronics and Telecommunication Engineering student with a strong interest in embedded systems, hardware design, automation, and control systems. I have hands-on experience with microcontrollers and development platforms including STM32, ESP32, ESP8266, PIC18F47K42, Arduino, and Raspberry Pi Pico, along with motor drivers, sensors, communication interfaces, and motion-control systems. I enjoy building, testing, and debugging hardware while understanding how embedded software interacts with electronic systems.

## Education

**Vishwakarma Institute of Technology, Pune** AUG 2023 – Present

- B.Tech in Electronics and Telecommunication Engineering
- CGPA: 8.92 / 10.0

**Higher Secondary (Class XII)** MARCH 2021 – 2023

- Maharashtra Technical Education Society Percentage: 78.83%

**Secondary (Class X)** JUNE 2015 – 2021

- Jnana Prabodhini Prashala, Pune Percentage: 92%

## Technical Skills

**Programming Languages:** Verilog, Python, C/C++      **Design:** Synopsys VCS/Verdi, Cadence Virtuoso  
**Tools:** Vivado, MPLAB IDE, Thonny, KiCAD      **Protocols:** UART, SPI, I2C, CAN

## Work Experience

**Embedded Intern – Aashman Technicals Pvt Ltd** JUN 2026 – AUG 2026

- Developed a 2-axis GRBL-based laser engraving system with 1 mm positioning accuracy. The system is intended for accurate agriculture, enabling weed detection using openCV and targeted laser burning to minimize crop damage.
- Designed and implemented a 7 pin input output level shifter using through hole and smd both components on kiCAD.

**Software Development Intern – Upthink Edutech Pvt Ltd** AUG 2024 – DEC 2025

- Developed and enhanced software features by writing clean, efficient code and collaborating with the team on real-world projects.
- Designed and implemented customized solutions to process, manage and integrate data efficiently based on project requirements.

**Electronics Head – International Society of Automation (ISA), VIT Pune** DEC 2023 – APR 2025

- Conducted a hands-on ROS workshop to train club members on robotics fundamentals; coordinated 3 sub-teams to deliver all project milestones on schedule

## Projects

**LoRa Based refrigerator control system (Sponsored by Amotech Lab)** AUG 2025 – DEC 2025

- Configured and validated UART communication on PIC18F452 for reliable LoRa module interfacing and data exchange.
- Successfully demonstrated a complete end-to-end LoRa-based monitoring and control system, showcasing strong skills in embedded system integration, wireless communication, and industrial automation system design

**UART Transmitter on FPGA** AUG 2025 – DEC 2025

- Design and integration of a basic UART (RX/TX) module on an FPGA to enable serial communication with external devices.
- Supports data transmission, reception, and simple debugging/monitoring through a reliable asynchronous serial interface.

## Relevant Coursework

- |                               |                                  |                       |
|-------------------------------|----------------------------------|-----------------------|
| • Digital Circuits            | • Git Github                     | • Raspberry Pi PICO   |
| • PIC18F474K42/452            | • C / C++                        | • STM32/ESP32/ESP8266 |
| • Robot Operating System(ROS) | • Data Structures and Algorithms | • Embedded Systems    |

## Awards Certificates and Publications

- |                                                                           |                                                      |
|---------------------------------------------------------------------------|------------------------------------------------------|
| • <b>IEEE Publication:</b> Smart Energy-Generating Keyboard               | • <b>2nd Place</b> – Modern College Hackathon (2024) |
| • <b>3rd Prize</b> – MLSC Hackspiration Hackathon                         | • <b>208th Rank</b> – Physics Brawl, Charles Univ.   |
| • <b>Udemy Course</b> – Verilog for an engineer with Xilinx vivado design | • <b>Maven Silicon</b> – Foundation Physical Design  |
| • <b>Udemy Course</b> – Advanced Computer Architecture and Organization   | • <b>Maven Silicon</b> – Digital Design              |